

Lan Zeng

Phone: (+86) 18024805066 | Email: lanzeng430@gmail.com | GitHub: github.com/LanZ430
The Chinese University of Hong Kong, Shatin, NT, Hong Kong SAR

RESEARCH STATEMENT

My research lies at the intersection of LLM-based agent systems, interactive narrative, multimodal generation, and human-computer interaction. I am particularly interested in how multi-agent orchestration can transform literary and cultural works into coherent, high-agency interactive experiences, while balancing user freedom, narrative consistency, long-form memory, and aesthetic fidelity. My prior work spans agent-driven interaction systems, retrieval-augmented generation, and human-centered evaluation in HCI settings.

EDUCATION

The Chinese University of Hong Kong, Hong Kong SAR

Sep. 2024 – Jun. 2025

M.S. in Information Engineering GPA: 3.33/4

Lancaster University College, England

Sep. 2020 – Jun. 2024

B.S. in Computer Science GPA: 3.55/4

Beijing Jiaotong University, China

Sep. 2020 – Jun. 2024

B.Eng. in Computer Science GPA: 3.5/4

RESEARCH EXPERIENCE

Research Assistant, AIoT Laboratory, The Chinese University of Hong Kong

Jun. 2025 – Present

Advisor: Prof. Guoliang Xing

- ContriCare (CHI 2026): Co-designed a community-based mutual aid platform for rural eldercare; conducted field studies and semi-structured interviews; led prototype development and qualitative analysis.
- ThermiKit (HotSense 2025): Led development of an edge-deployable thermal analytics framework with agent-driven natural-language interaction over sensor data.
- Co-developed an AI-assisted eldercare system integrating thermal sensing, fall detection, and agent-based care workflows for privacy-sensitive monitoring environments.

Research Assistant (Part-time), Institute of Software, Chinese Academy of Sciences

Dec. 2023 – Apr. 2025

Advisor: Kaichun Yao

- Fine-tuned multiple Chinese open-source LLMs (ChatGLM, Chinese-Vicuna, Baichuan2) on large-scale technical QA data and developed a retrieval-enhanced question answering system leveraging online knowledge sources.
- Conducted research on knowledge-aware follow-up query generation and co-developed R²AG, an end-to-end framework for progressive information acquisition through adaptive questioning.
- Co-developed SciPG, a multimodal extractor-generator framework for automatic scientific poster generation from academic papers, including benchmark construction and model evaluation.

PUBLICATIONS

1. Guo, Y., **Zeng, L.**, He, L., et al. (2026). Beyond Transactions: Sustaining Rural Mutual Aid through Mediated Reciprocity. CHI 2026 Extended Abstracts (Poster), Article 127, 1–5. DOI: 10.1145/3772363.3799061
2. Wang, G., **Zeng, L.**, Chen, L.-K., et al. (2026). Noise & Interference Loading for Capacity on Demand with Fine-Tuned LLM-Assisted Optical Access Network Automation. OECC 2026. (Student Travel Grant Awardee)
3. **Zeng, L.**, Huang, C., Xie, R., et al. (2025). ThermiKit: Edge-Optimized LWIR Analytics with Agent-Driven Interactions. Proceedings of the 2025 ACM International Workshop on Thermal Sensing and Computing (HotSense 2025), 40–46. DOI: 10.1145/3737905.3769284
4. **Zeng, L.**, Yao, K., Qin, C., et al. (2024). SciPG: A New Benchmark and Approach for Layout-Aware Scientific Poster Generation. OpenReview.
5. Yao, K., **Zeng, L.**, Qin, C., et al. (2024). What to Ask LLM: Knowledge-Aware Follow-Up Query Generation for Progressive Information Acquisition. OpenReview.

PROJECTS

DC-HMAS – Literary-to-Interactive Narrative System

Independent Research, Present

- Designed and implemented a TRPG-inspired prototype for adapting literary texts into interactive narrative experiences, featuring hierarchical memory, dual-script design, and Host Agent orchestration.
- Iterating on the system through research discussions with Prof. Jussi Holopainen (City University of Hong Kong) on prompt design, narrative control, and character-goal guidance.
- GitHub link <https://github.com/LanZ430/DC-HMAS-Demo>

NovelVerse – Multi-Modal AI Agent for Immersive Fictional Interaction

Solo Project, 2025

- Designed and developed an agent-based interactive storytelling system that integrates LLMs and image generation APIs to create first-person, multi-modal narrative experiences. The system features role-based interaction, story deviation evaluation, cross-chapter memory, and a Streamlit web UI, enabling dynamic text-to-image storytelling for user-driven world exploration.
- GitHub link <https://github.com/LanZ430/NovelVerse>

Blind-Friendly Button Assistance System

Human-Computer Interaction Course Project, 2023

- Built a tangible prototype to assist visually impaired users in identifying household appliance buttons using audio feedback, infrared sensing, and tactile cues. Led the interaction and hardware design, and conducted user studies showing a significant reduction in task completion time ($p < 0.05$).

Cyberpunk 3D Exploration Game (Unreal Engine 4)

Team Project, 2022

- Created a large-scale third-person 3D exploration game integrating shooting, puzzle-solving, and racing mechanics within a cyberpunk world. Designed new interactive scenes, integrated AI dialogue modules, and implemented multi-mode player interactions to enhance engagement.

AWARDS

- May. 2023 Beijing Municipal Recognition, National College Students' Innovation and Entrepreneurship Training Program
- Apr. 2023 Second prize in Campus Selection, The 25th “FLTRP. ETIC Cup” National English Debating Competition
- Oct. 2022 Provincial First Prize in Contemporary Undergraduate Mathematical Contest in Modeling
- Apr. 2022 Honorable Mention of International Mathematical Contest in Modeling
- Nov. 2021 Provincial Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant

Sep. 2021 – Jan. 2022

BJTU Communication Engineering C Language Course

Shandong, China

- Evaluated 80 students' programming assignments across 16 groups in the C Language course.
- Provided the teacher with suggestions for the next homework according to the overall feedback.
- Supported the instructor by designing slides and 10 sample problems per class to clarify syntax and algorithm concepts.

Calculus Lecturer

Sep. 2020 – Jul. 2021

Student Academic Support Centre, Beijing Jiaotong University

Shandong, China

- Assisted in teaching Calculus to over 200 first-year undergraduates, covering differentiation and multiple integrals.
- Conducted weekly review workshops with 20 + students per session, providing detailed problem-solving guidance.
- Prepared lecture materials and exam-week review sessions to strengthen students' understanding of core topics.